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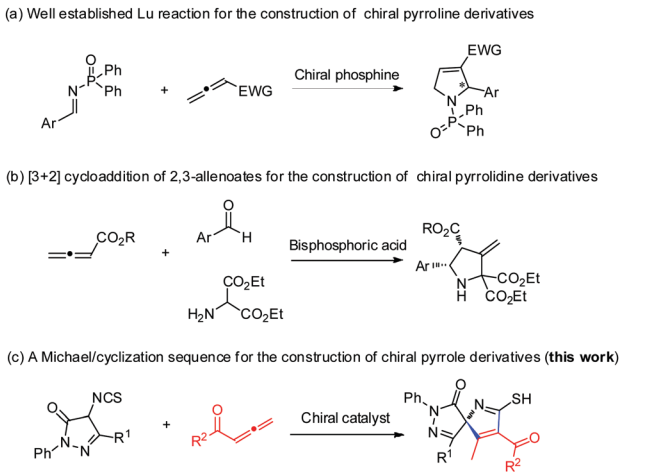
Asymmetric sequential annulation/aldol process of 4-isothiocyanato pyrazolones and allenones: access to novel spiro[pyrrole–pyrazolones] and spiro[thiopyranopyrrole–pyrazolones]†

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A catalytic asymmetric sequential annulation/aldol reaction of 4-isothiocyanato pyrazolones and allenyl ketones has been developed, which furnished a series of spiro[pyrrole–pyrazolone] heterocycles and structurally novel spiro[thiopyranopyrrole–pyrazolone] derivatives in good yields with high to excellent enantioselectivities. Notably, parallel resolution of racemic spiro[pyrrole–pyrazolones] was achieved by a catalyst-controlled asymmetric intramolecular vinylogous aldol process. Structure diversity of the product was further enhanced by ready transformations.

Nitrogen-containing five-membered heterocycles are ubiquitous motifs found in various natural products, pharmaceutical agents, agrochemicals and synthetic materials.[1] Accordingly, the development of new efficient methods toward their construction has long been a sought-after research topic in synthetic organic chemistry.[2] Within this context, 2,3-allenyl carbonyl compounds have emerged as a versatile class of building block for the construction of chiral pyrrolidine derivatives.[3] In this regard, seminal contributions from the Lu group are of particular note, who reported the [3+2] cyclization process between 2,3-allenoates and imines with phosphine catalysis, leading to the formation of pyrroline rings with the allenoate being a three-carbon component.[4] Later, the asymmetric version of this reaction was extensively explored by using various chiral phosphine or amine-based nucleophilic catalysts (Scheme 1a).[5] Other than working as a three-carbon synthon, 2,3-allenyl carbonyl compounds as a two-carbon partner were also engaged in the asymmetric construction of pyrrolidine derivatives, as exemplified by the chiral phosphoric acid catalyzed [3+2] cycloaddition between azomethine ylides and 2,3-allenoates pioneered by Gong and co-workers (Scheme 1b).[6] However, compared to the well-established Lu reaction, the synthetic potential of 2,3-allenyl carbonyls as two-carbon synthons toward pyrrolidine/pyrrole construction is much less explored,[7] and effort toward this end is thus necessary and interesting in the context of new reaction design.

As another kind of important five-membered nitrogen heterocycle containing two adjacent nitrogen atoms, pyrazolones constitute a notable pharmacophore seen in an array of pharmaceutical molecules and bioactive compounds.[8] And the asymmetric derivatization of pyrazolones has become a topic of intense recent research.[2h,n] In continuation of our program toward the development of new asymmetric reactions for accessing chiral nitrogen heterocycles of medicinal relevance, we very recently developed a reactive ambiphilic pyrazolone synthon by introducing an electrophilic isothiocyanato group into the nucleophilic C4 position of the pyrazolone ring, which showed excellent reactivity toward imines[9] and activated alkenes[10] leading to novel chiral spiropyrazolone



Scheme 1 Previous synthesis of chiral pyrrole derivatives with allenoates and our asymmetric strategy.

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† Electronic supplementary information (ESI) available: Experimental procedures and detailed characterization data of all new compounds (PDF). CCDC X-ray crystal details for **5af** (CIF) 2034086 and **5′′af** (CIF) 2040382. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d0cc07113g

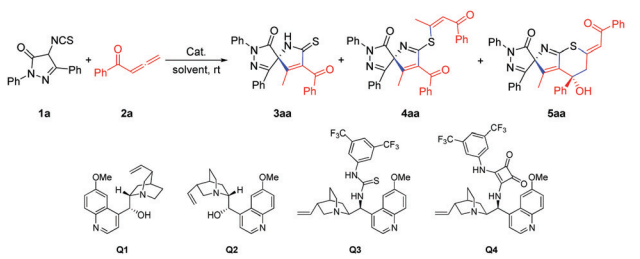
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products *via* organocatalyzed cascade spiroannulation. We further wondered whether 2,3-allenyl carbonyl compounds could be engaged as a two-carbon component in an asymmetric annulation with the 4-isothiocyanato pyrazolone synthon to forge a chiral spiro 2*H*-pyrrole scaffold (Scheme 1c). Herein, we report our successful realization of this assumption.

We commenced our studies by surveying the reaction of 4-isothiocyanato pyrazolone **1a** and phenyl allenone **2a** in the presence of cinchona alkaloid catalysts (Table 1). An initial trial revealed that the reaction could hardly stop at the product **3aa**, but it instead continued to add to another molecule of phenyl allenone **2a** to give the product **4aa** together with the intramolecular aldol cyclization product **5aa**. In order to ensure the full conversion of **1a**, in the following screening experiments 2.5 equivalent of **2a** was used.

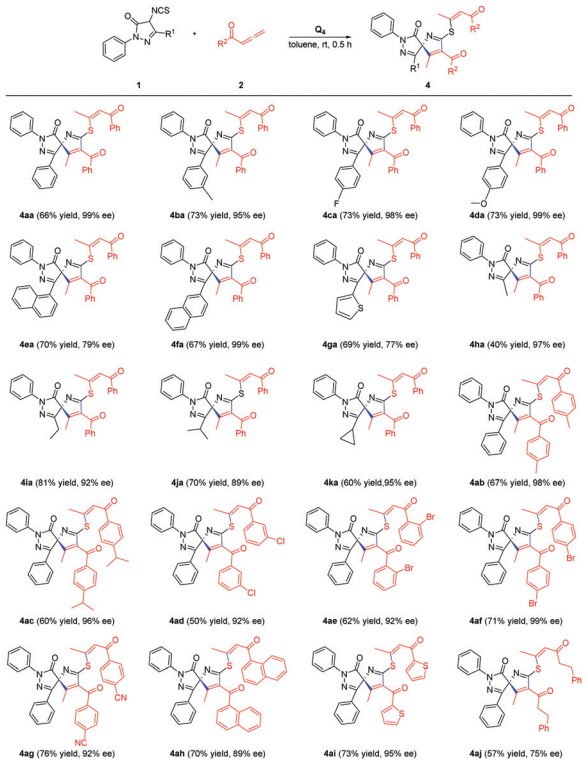
Natural quinine (**Q1**) was first examined as the catalyst, and the reaction proceeded rapidly in dichloromethane (DCM) to give the cascade spiroannulation/sulfa-Michael addition product **4aa** in 59% yield and 31% ee, along with the formation of aldol cyclization product **5aa** in 20% yield (entry 1). A similar product distribution was observed when quinidine (**Q2**) was used but with decreased ee of 11% favoring the antipode of **4aa** (entry 2). Next, quinine-derived hydrogen bonding catalysts were examined. While quinine thiourea **Q3** afforded **4aa** in 59% yield and 52% ee (entry 3), to our delight, the use of quinine squaramide catalyst **Q4** improved the enantioselectivity to 91% ee. Then, using **Q4** as the optimal catalyst, a series of solvents were screened (entries 5–9). The result indicated that

Table 1 Optimization of the reaction conditions



Entry	Cat.	Solvent	t [h]	Yield[a] [%] 3aa	4aa	5aa	Ee[b] [%] of 4aa
1	**Q1**	DCM	0.5	Trace	59	20	31
2	**Q2**	DCM	0.5	Trace	57	19	−11
3	**Q3**	DCM	0.5	Trace	59	18	52
4	**Q4**	DCM	0.5	Trace	59	15	91
5	**Q4**	$CHCl_3$	0.5	Trace	60	19	96
6	**Q4**	THF	0.5	Trace	66	18	94
7	**Q4**	Toluene	0.5	Trace	66	18	99
8	**Q4**	CH_3CN	0.5	Trace	57	20	54
9	**Q4**	DCE	0.5	Trace	60	15	93
10[c]	**Q4**	Toluene	0.5	Trace	66	15	97
11[d]	**Q4**	Toluene	0.5	Trace	70	13	82
12	—	Toluene	6	Trace	80	Trace	0

The reaction was performed on a 0.1 mmol scale with **1a** (1.0 equiv.), **2a** (2.5 equiv.), and catalyst (10 mol%) in solvent (1.0 mL). [a] Isolated yield. [b] Determined by chiral HPLC analysis. [c] Cat. (5 mol%). [d] Cat. (2.5 mol%).



Scheme 2 Substrate scope investigation. The reaction was carried out on a 0.2 mmol scale with **Q4** (10 mol%) in 2.0 mL toluene; the molar ratio of **1** and **2** was 1 : 2.5. Isolated yield was given. The ee was determined by chiral HPLC.

except for acetonitrile, other solvents including chloroform ($CHCl_3$), tetrahydrofuran (THF) and 1,2-dichloroethane (DCE) showed better enantioselectivities than DCM. Pleasingly, when the reaction was run in toluene, the ee value could be increased to 99% with a 66% yield of **4aa** (entry 7). Given the extremely high efficiency of **Q4**, reactions with lower catalyst loadings were investigated. Thus, with 5 or 2.5 mol% **Q4**, still a very fast reaction was observed, without any loss of the product yield but with decreased ee values to 97% and 82%, respectively (entries 10 and 11). Notably, without any catalyst, this reaction also proceeded smoothly to completion within 6 h and **4aa** was formed in 80% yield in racemic form.

With the optimal conditions determined, we investigated the generality of this new cascade reaction. A diverse array of the 4-isothiocyanato pyrazolones was first surveyed. As demonstrated in Scheme 2, pyrazolones with various substituents on the phenyl ring at the C3 position of the pyrazolone framework, as exemplified by methyl, fluorine, and methoxyl groups were smoothly converted to the corresponding products in good yields with excellent enantioselectivities of 93–99% ee (**4ba–4da**). The polycyclic aromatic and heteroaromatic substituents at the C3 position of the pyrazolones were also compatible under standard reaction conditions, generating the desired product **4ea**, **4fa** and **4ga** in good yields with 79%, 99% and 77% ee, respectively. In addition, the alkyl groups at the C3 position were also accommodated, as the methyl, ethyl, isopropyl and cyclopropyl substituted pyrazolones furnished the products **4ha**, **4ia**, **4ja** and **4ka** in 40–81% yield with 89–97% ee.

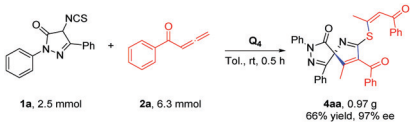
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Next, the substrate scope with respect to the allenyl ketone partner was evaluated. To our delight, all tested 1-aryl allenic ketones with electron-donating and electron-withdrawing groups such as methyl, isopropyl, halogen and cyano groups at the *para*-, *meta*- or *ortho*-positions were well accommodated in this process, affording the corresponding products (**4ab–4ag**) in good yields with excellent enantioselectivities (92–99% ee). Notably, 1-naphthyl and 2-thienyl allenic ketones also generated the desired products **4ah** and **4ai** in good yields with excellent enantioselectivities of 89% ee and 95% ee. Besides, the aliphatic allenyl ketone was well accommodated (**4aj**). The geometry of the olefinic double bond of the side chain was determined to be *Z* by NOE study of **4af** (for details see the ESI†).

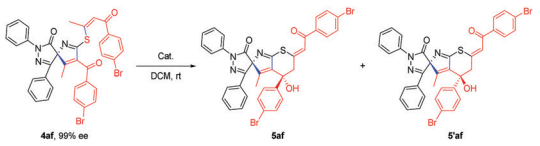
To evaluate the practical utility of the current catalytic system, a scale-up reaction was performed with 4-isothiocyanato pyrazolone **1a** (2.5 mmol), phenyl allenone **2a** (6.3 mmol), and **Q4** (0.25 mmol), and the cascade spiroannulation/sulfa-Michael addition was achieved smoothly to afford **4aa** in 66% yield with 97% ee on a gram-scale (Scheme 3). Further transformations of the products were also conducted (see the ESI,† for details).

It is worth noting that the polyheterocyclic product **5aa** observed in the condition optimization stage is structurally novel and interesting, which could actually be obtained in 68% yield by prolonging the reaction time to 36 h and presumably ought to be derived from compound **4aa** through an intramolecular vinylogous aldol reaction. Given the marked structural novelty, as demonstrated by both spiro and fused ring patterns featuring pyrazolone, 2*H*-pyrrole, and thiopyran frameworks in **5aa**, we set out to investigate the construction of **5** after the effective synthetic protocol to **4** was established. Hence, as summarized in Table 2, when DABCO or Et_3N was used for **4af** as a model substrate, the aldol cyclization product **5af** and **5′af** could be obtained in good total yield but with low dr (entries 1 and 2). We were pleased to find that **Q4** furnished improved activity and selectivity, and the corresponding spiro[thiopyranopyrrole–pyrazolone] **5af** was furnished in high yield with excellent dr and ee values (entry 3). It turned out that the diastereoselectivity of the reaction is controlled by the chiral catalyst, and the spirostereocenter in **4af** has no directing effect, which was different from Evans aldol additions.[11] Then, **Q4** catalyzed asymmetric vinylogous aldol reaction of various **4** was investigated. As shown in Scheme 4, the yield and stereoselectivity could be sustained at a very high level for a wide variety of pyrazolones substituted with different groups at the C3 position, affording a series of chiral fused-thiopyran scaffolds, an important core of a large family of medicinally relevant compounds.[12] Furthermore, the absolute configuration of **5af** was determined by X-ray crystallographic analysis.



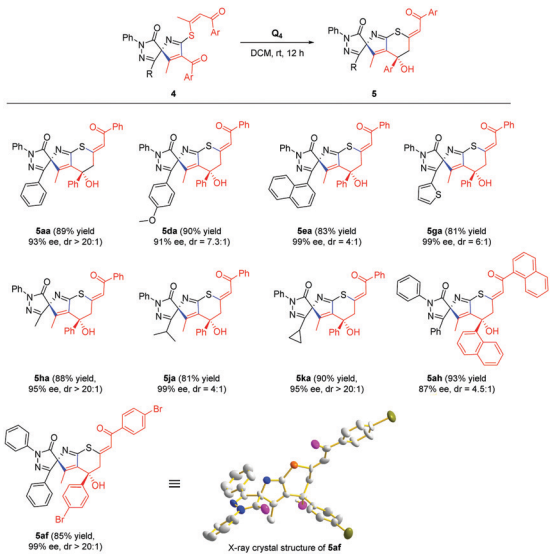
Scheme 3 Gram-scale reaction.

Table 2 Optimization of the asymmetric vinylogous aldol reaction



Entry	Catalyst	t [h]	Dr	Yield[a] [%] 5af	Yield[a] [%] 5′af	Ee[b] [%] 5af	Ee[b] [%] 5′af
1	DABCO	24	1 : 3	14	31	99	99
2	Et_3N	24	1 : 1.4	27	33	99	99
3	**Q4**	12	>20 : 1	85	—	99	—

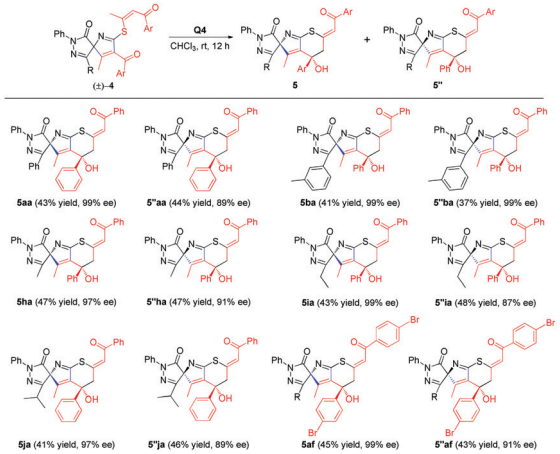
The reaction was carried out on a 0.1 mmol scale in 1.0 mL DCM with catalyst (10 mol%). [a] Isolated yield was given. [b] The ee was determined by chiral HPLC. The dr was determined by ^{1}H NMR of the crude reaction mixture.



Scheme 4 Synthesis of spiro[thiopyranopyrrole–pyrazolone] derivatives **5** by asymmetric vinylogous aldol reaction of **4**. The reaction was carried out on a 0.1 mmol scale in 1.0 mL DCM with **Q4** (10 mol%). Isolated yield of the total was given. The dr was determined by ^{1}H NMR of the crude reaction mixture. The ee was determined by chiral HPLC.

Inspired by the success of the above catalyst-controlled asymmetric aldol cyclization, the parallel resolution of racemic compounds **4** to enantioenriched products **5** catalyzed by cinchona alkaloids was next conducted (see Table S1 in the ESI,† for details). To our delight, **Q4** exhibited superior reactivity and stereocontrol outcome affording the aldol product **5af** within 12 h in a 45% yield with 99% ee, and the product **5″af** in 43% yield with 91% ee which is the enantiomer of **5′af**, and the absolute configuration of **5″af** was determined by X-ray crystallographic analysis (see ESI,† for details). And a series of diastereomer pairs of spiro[thiopyranopyrrole–pyrazolones] under the optimal reaction conditions were obtained in good yield with excellent enantioselectivity (Scheme 5).

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Scheme 5 Synthesis of spiro[thiopyranopyrrole–pyrazolone] derivatives **5** by asymmetric vinylogous aldol of (±)-**4**. The reaction was carried out on a 0.1 mmol scale in 1.0 mL $CHCl_3$ with **Q4** (10 mol%). Isolated yield was given. The ee was determined by chiral HPLC.

In conclusion, we have developed an efficient and facile catalytic asymmetric [3+2] annulation of 4-isothiocyanato pyrazolones and allenyl ketones. This reaction afforded spiro[pyrrole–pyrazolone] products with excellent enantio-selectivities with a broad substrate scope. And notably, a further catalyst-controlled asymmetric vinylogous aldol cyclization delivered the spiro[thiopyranopyrrole–pyrazolone] derivatives readily in good yields with excellent stereoselectivities. Racemic spiro[pyrrole–pyrazolones] could also be converted into enantioenriched spiro[thiopyranopyrrole–pyrazolone] diastereomer pairs.

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Conflicts of interest

There are no conflicts to declare.

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